

Carlos Alcala

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EDUCATION

Loyola University, Chicago, IL
B.S. Software Engineering — Expected Graduation: **December 2027**

TECHNICAL SKILLS

Languages: Swift, Java, Python, JavaScript, Racket, C++
Technologies & Tools: Firebase, SQL/F1, gRPC, CoreML, OpenCV, Waymax, Twilio, Xcode, GitHub, IntelliJ, SolidWorks
Skills: Backend Development, Machine Learning, Computer Vision, Databases, CAD, Kinematic Modeling, iOS Dev, MVVM arch

PROFESSIONAL EXPERIENCE

Google Software Engineering Intern (ASDI/STEP), Sunnyvale, CA — May 2025 - August 2025

- Streamlined client-targeting workflows for internal teams by developing an analytics page that retrieved and visualized live promotion data, including clicks, views, and reach.
- Handled high-traffic requests and enabled real-time updates by engineering asynchronous, gRPC-based server calls to fetch live data from the F1 promotions SQL database.

EveryoneCanCode Apple App Development Instructor, Chicago, IL — Jun 2024 - Aug 2024, Jun 2026 - Aug 2026

- Guided 50+ students through the full iterative software lifecycle by instructing Swift app development best practices.
- Mentored four student teams to place in the Top 4 of Apple's citywide app competition by facilitating the integration of advanced backend frameworks (Firebase, Twilio, HealthKit, and CoreML) to solve real-world problems.

Apple App Accelerator + Entrepreneurship program, Chicago, IL — June 2023 - December 2023
Software Engineering Intern

- Developed an iOS system using Swift and pitched as 1 of only 4 citywide finalist teams selected from a highly competitive incubator program at the Chicago Downtown Apple Store.
- Optimized real-time hospital alerts by building a robust Firebase database and priority-filtering logic for **Kalcom**, a call-light iOS application to help patients receive prioritized attention efficiently.

PROJECTS & LEADERSHIP

Motorcycle Decision-Making Simulation for Emergency Cases | Personal Project — **In Progress**

- Modeling a motorcycle's kinematics and braking limits from published rider-skill studies
- Designing a brake vs acceleration decision making logic for motorcycle physics: slower braking, smaller area, faster acceleration.
- Built a custom motorcycle agent and integrated it into Waymo's Waymax simulator using the Open Motion Dataset.

Retargeting Kinematics Hand | Research — Apr 2026 - Jun 2026

- Achieved high-confidence 3D hand tracking and motion prediction by building a real-time computer vision pipeline utilizing MediaPipe along with the **PyRoki** kinematic optimization toolkit to translate hand gestures to the robotic hand in real-time.
- Improved replication accuracy by implementing a finger-weighted kinematic cost function in the Levenberg-Marquardt solver.

LIGHT Engineering Team Lead — Nov 2025 -Current

- **Founded** Loyola University Chicago's first dedicated engineering project team, creating an environment for students across majors to gain hands-on experience in design, manufacturing, and programming.
- Led multidisciplinary development of an electric vehicle for Purdue's evGrandPrix, coordinating mechanical, electrical, and software teams through CAD design, fabrication, system integration, and testing.

NU Solar Car Team - Mechanical Subteam — Sept 2025 - Current

- Joined NU Solar Car Team as its only non-Northwestern student, contributing to its solar-powered vehicle.
- Designed brake pedals, suspension components, and rotors in CAD, validated designs through Finite Element Analysis, and machined parts in the Ford Engineering Design Lab.

Sheltie - Natural Disaster Model | Personal Project — Nov 2024 - Feb 2025

- Developed a real-time structural hazard detection system by training a CoreML computer vision model on structural fatigue datasets and deploying it in an iOS app for live camera-based risk detection and safety guidance.

AWARDS

Engineering: Congressional App Challenge Winner | Top 4 Apps EveryoneCanCode+Create | National Recognition Program Awardee